Weekly Project Report - Week 1

Web-Based Certificate Scanning and Error Detection System

1. Tasks Completed, In Progress, and Planned

Status	Task	Progress
Completed	Project requirements and problem understanding	Completed
Completed	Initial analysis of scanned certificate verification	Completed
Completed	Decision to drop pixel-based comparison and verify certificates using OCR (Hindi and English) only	Completed
Completed	Basic frontend prototype (mock data, basic pages only), built in 5 parts: (1) app shell, navigation and responsive layout; (2) dashboard; (3) upload and verification progress; (4) comparison viewer and field results; (5) verification result and human review	Completed
Completed	Testing of the basic prototype on Windows and for mobile responsiveness	Completed
In Progress	Evaluation of OCR for Hindi and English text	In progress
In Progress	Research on image preprocessing (alignment, noise and lighting normalization) to improve OCR accuracy	In progress
Planned	Backend (FastAPI) and integration with the frontend	Planned
Planned	OCR pipeline prototype (Tesseract) for extracting certificate fields	Planned
Planned	Field-level matching against reference data (RapidFuzz)	Planned
Planned	Human-assisted verification workflow connected to the backend	Planned

2. Problems

- Hindi and English text are present in the certificates, making OCR-based verification more challenging.
- Scanned documents may contain rotation, scaling, uneven lighting, blur, compression artifacts, and scanner noise, which can reduce OCR accuracy and need preprocessing.
- The certificate verification process has a zero-tolerance requirement, so automated detection should assist a human reviewer rather than be treated as the final authority.
- Further investigation is required to distinguish genuine printing/layout errors from OCR misreads caused by scanning conditions.
- Only a basic frontend prototype is complete and it runs on simulated (mock) data; backend integration is pending, so end-to-end behavior has not been validated.

3. Time Spent by Team Members

Team Member	Task	Hours
Member 1	Project requirement analysis	1.5
Member 1	Frontend part 2: Dashboard and mock data	1
Member 2	OCR / Hindi text research	2
Member 2	Frontend part 3: Upload Certificate and Verification Progress pages	1
Member 3	Image-processing research	1
Member 3	Frontend part 5: Verification Result and Human Review pages	1
Member 4	Frontend part 4: Comparison Viewer and Field Results pages	1.5
Member 5	Backend / system research	1
Member 5	Frontend part 1: App shell, sidebar navigation and responsive layout	1.5
Total		**11.5**

Frontend hours include testing of the respective part on Windows and mobile.

4. Last Meeting with the PoC

Date: 31 - Aug - 2026
Discussion: About Hindi and English text in certificates, zero error tolerance, verification of documents, heavily time-consuming manual process and human-assisted verification workflow.

5. Date of Current Build

Build date: 24 - Sep - 2026 (basic frontend prototype with mock data; backend not yet integrated)

6. Results of Tests on the Build

Metric	Result
Total number of tests	2
Successful tests	2
Unsuccessful tests	0

Tests performed: (1) Windows desktop browser - layout and navigation across all pages; (2) mobile responsiveness - layout and sidebar behavior on a small screen.

7. Defects Found in QA Process

No defects were found in the Windows and mobile responsiveness tests. Functional QA of OCR and verification logic has not been conducted yet, as the backend is pending.